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The ML Classroom

Time Series Analysis and Forecasting with ARIMA in Python

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81

Time series forecasting is a crucial area of machine learning that predicts future points in a series based on past data. It is especially common in economics, weather forecasting, and capacity planning, but it is applicable in many other fields. Today, we'll walk through an example of time series analysis and forecasting using the ARIMA model in Python.

Understanding ARIMA

ARIMA stands for AutoRegressive Integrated Moving Average. It is a class of models that captures a suite of different standard temporal structures in time series data. Before we apply ARIMA, we need to ensure that the series is stationary, which involves checking if its statistical properties such as mean, variance, and autocorrelation are constant over time.

The Data

We are using a dataset that represents the daily total female births in California in 1959, which is commonly used as an example in time series analysis.

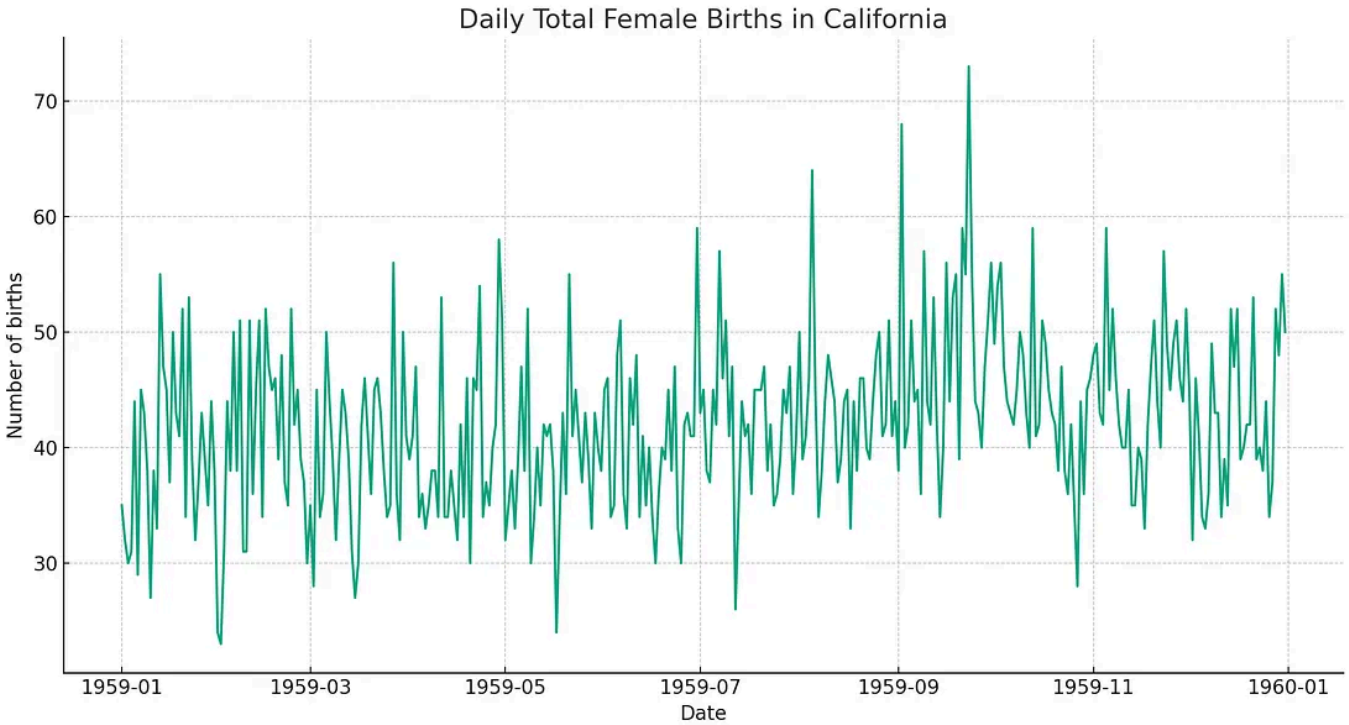
Preliminary Analysis

We start by loading the data and plotting the time series:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
data = pd.read_csv('daily-total-female-births-CA.csv')
data['date'] = pd.to_datetime(data['date'])
data.set_index('date', inplace=True)
# Plot the time series
plt.plot(data['births'])
plt.title('Daily Total Female Births in California')
plt.xlabel('Date')
plt.ylabel('Number of births')
plt.show()
```

The plot helps us visualize the data and provides initial insights into its properties, like trends and seasonality.



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Testing for Stationarity

Next, we use the Augmented Dickey-Fuller (ADF) test to check for stationarity:

```
from statsmodels.tsa.stattools import adfuller

adf_test = adfuller(data['births'])
# Output the results
print('ADF Statistic: %f' % adf_test[0])
print('p-value: %f' % adf_test[1])
```

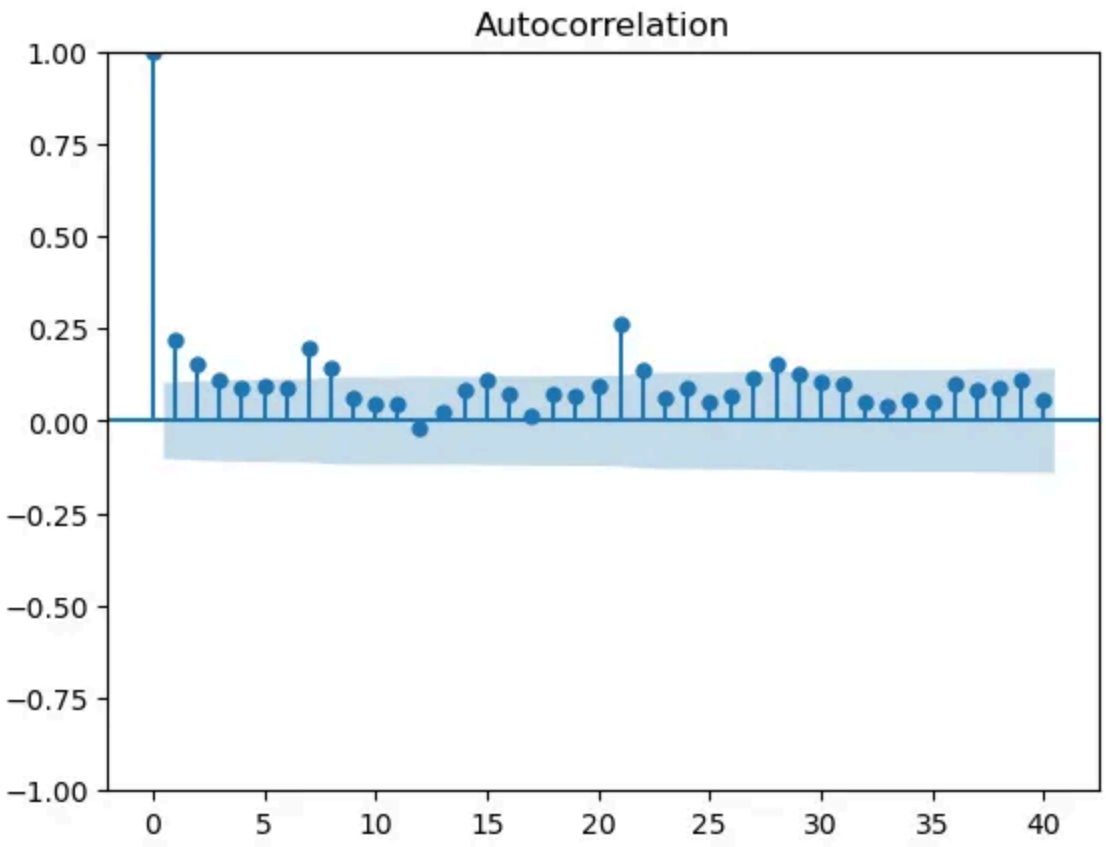
```
ADF Statistic: -4.808291
p-value: 0.000052
```

A p-value below 0.05 indicates stationarity, and our data meets this criterion, so we do not need to difference it.

Finding ARIMA Parameters

We use the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots to find the ARIMA parameters.

```
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
plot_acf(data['births'], lags=40)
plot_pacf(data['births'], lags=40)
plt.show()
```



These plots suggest that an ARIMA(1, 0, 1) model may be a good starting point.

Building the ARIMA Model

```
from statsmodels.tsa.arima.model import ARIMA

model = ARIMA(data['births'], order=(1, 0, 1))
model_fit = model.fit()
```

Training and Forecasting

We train the model on the data and perform a forecast.

```
forecast = model_fit.get_forecast(steps=30)
```

Then we visualize the forecast alongside the historical data.

Model Evaluation

To assess the model, we perform a retrospective forecast.

```
from sklearn.metrics import mean_squared_error

# Split the data into train and test
train_size = int(len(data) * 0.8)
train, test = data[0:train_size], data[train_size:len(data)]

# Fit the ARIMA model on the training dataset
```

```
model_train = ARIMA(train['births'], order=(1, 0, 1))
model_train_fit = model_train.fit()

# Forecast on the test dataset
test_forecast = model_train_fit.get_forecast(steps=len(test))
test_forecast_series = pd.Series(test_forecast.predicted_mean, index=test.index)

# Calculate the mean squared error
mse = mean_squared_error(test['births'], test_forecast_series)
rmse = mse**0.5

# Create a plot to compare the forecast with the actual test data
plt.figure(figsize=(14,7))
plt.plot(train['births'], label='Training Data')
plt.plot(test['births'], label='Actual Data', color='orange')
plt.plot(test_forecast_series, label='Forecasted Data', color='green')
plt.fill_between(test.index,
                 test_forecast.conf_int().iloc[:, 0],
                 test_forecast.conf_int().iloc[:, 1],
                 color='k', alpha=.15)
plt.title('ARIMA Model Evaluation')
plt.xlabel('Date')
plt.ylabel('Number of Births')
plt.legend()
plt.show()

print('RMSE:', rmse)
```

```
RMSE:  6.970853456222879
```